MIL-S-52866B
25 April 1979
SUPERSEDING
MIL-S-52866A(ME)
15 March 1977

MILITARY SPECIFICATION

SENSOR, MAGNETIC WEAPON

DT-547/FSS-9(V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Magnetic Weapon Sensor, which is used to detect the magnetic field disturbance caused by the removal of a weapon from a weapons rack. It is a component of the Joint-Services Interior Intrusion Detection System (J-SIIDS).

2 APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Locked corners.
PPP-B-636	- Boxes, Shipping, Fiberboard.

FSC 6350

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-676 - Boxes, Setup.

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MIL-P-116 - Preservation-Packaging, Methods of.
MIL-T-704 - Treatment and Painting of Material.
MIL-C-22751 - Coating System, Epoxy Polyamide, Chemical Solvent Resistant: Processes for Application of.
MIL-C-45662 - Calibration System Requirements.

STANDARDS

FEDERAL

FED. STD. No. 595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests: Expotential Distribution.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1188 - Commerical Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13220E2900 - Sensor, Magnetic Weapon - DT-547/FSS-9(V).
TA13220E2899 - Toroid Assembly.

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TA13220E2898	- Input and Filter Assembly.
TA13220E2894	- Circuit Card Assembly, Signal Processor.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Magnetic Weapons Sensor (hereinafter called MWS) shall be in accordance with Top Assembly TA13220E2900, and as specified herein.

3.1.1 Major components. The major components of the MWS are as follows:

(a) Loop wire (16 AWG).
(b) Toroid Assembly TA13220E2899.
(c) Signal Processor Enclosure.
(d) Input and Filter Assembly TA13220E2898.
(e) Circuit Card Assembly, Signal Processor TA13220E2894.

When specified (see 6.2), any of the above major components shall be furnished separately.

3.1.2 Function. The MWS is intended to detect unauthorized removal of weapons from arms racks. The loop wire transducer shall be easily adaptable to different arms rack configurations. The number of racks that shall be protected is limited only by the length of the 16 AWG wire. The movement of a weapon induces a current in the loop which is fed to the signal processor where the alarm is generated. The signal processor shall consist of a power supply, low noise band-pass amplifier, and bipolar threshold detector. The rejection of extraneous noise (false alarm) (see 6.4.1), is based on the loop configuration and band-pass characteristics of the amplifier.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits,

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the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the MWS or major components (when procured separately). Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more MWS's or major components (when procured separately) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce MWS's or major components (when procured separately) that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4. Unless otherwise specified, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more MWS's for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.8).

3.5 Environmental conditions. The MWS or major components (when procured separately) shall withstand individual or combinations of the environmental conditions specified herein without damage (see 6.4.2), or performance degradation.

3.5.1 Operational.

(a) Temperature range: -20° F to +150° F.
(b) Humidity up to 95 percent relative.

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3.5.2 Nonoperational and storage.

(a) Temperature range: -30° F to +165° F.
(b) Humidity: up to 95 percent relative.
(c) Shock: 20 times the acceleration due to gravity (g's) for 11 milliseconds (ms) duration and bench handling.
(d) Vibration: transportation conditions.

3.6 Detailed technical requirements.

3.6.1 Tamper Protection. The inpedance between Terminals 5 and 6 of TB1 shall represent a tamper alarm condition by exceeding 100 kilohms (k ohms) or a nontamper condition by being under 10 ohms. The tamper alarm condition shall be presented whenever the signal processor cover is opened from 0.062 to 0.187 inch and shall continue until the cover is closed or the tamper alarm plunger is pulled outward. The alarm shall reset by closing the cover.

3.6.2 Terminal board (TB) connections.

TB1	Function
1	Alarm Contact
2	Alarm Contact
3	Spare
4	Spare
5	Tamper alarm Contact
6	Tamper alarm Contact
7	+20 Volts direct current (Vdc)
8	20 Vdc Return
9	Enclosure Ground
10	Spare

On TB1, removable jumpers shall be provided across Terminals 2 and 3, 3 and 4, and 4 and 5.

3.6.3 Functional Characteristics.

3.6.3.1 Power. The MWS shall function on regulated voltages between 18 and 22 Vdc. The current requirements shall not exceed 60 milliamperes (mA). Reverse voltage protection shall be incorporated in the MWS.

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3.6.3.2 Alarm input sensitivity. The signal processor shall alarm on power interruption exceeding 0.5 seconds and an input voltage of 25 to 100 nanovolts peak ±15%, at frequency of 0.65 Hertz (Hz), of any polarity. The alarm level shall be adjusted by a screwdriver-adjustable potentiometer, with number set positions.

3.6.3.3 Recovery. After an overload signal (saturation of the final analog amplifier chain) or removal or reconnection of the loop wire, the amplifier shall process a normal signal within 15 seconds after the condition. After receiving an alarm input, the signal processor shall present alarm conditions for not less than 0.5 second nor more than 5.0 seconds after the alarm input is removed.

3.6.3.4 Circuit performance. The signal processor input noise shall be less than 4 nanovolts during a 60 second period with a 3 ohm load across the input.

3.6.3.5. Band-pass characteristics. The 4 decibel (dB) (±2.0 dB) band-pass limits shall be 0.35 to 1.3 Hz. There shall be only one inflection point in the 4 dB band-pass. The gain shall be down at least 35 dB at 10 Hz and shall decrease at the rate of 30 dB per octave or greater above 10 Hz. The gain shall be down at least 20 dB at 0.1 Hz and shall decrease at the rate of 12 dB per octave or greater below 0.1 Hz.

3.6.3.6 Intermodulation. The MWS shall not alarm when two signals of 32 Hz and 30 Hz, both at an amplitude of 150 microvolts root mean square (rms), are applied to the input at the same time. The MWS shall not alarm when two signals of 60 Hz and 62 Hz, both at an amplitude of 350 microvolts rms, are applied to the input at the same time.

3.6.3.7 Alarm conditions. The alarm output conditions across the signal processor contacts (TB1-1 and TB-2) shall be as follows:

No Alarm - 0 to 10 ohms
Alarm - Greater than 100k ohms

The alarm circuit shall accept up to 10 mA of direct current in the no-alarm state, and up to 30 Vdc in the alarm state.

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3.6.4 Toroid assembly. The toroid assembly (end of line module) shall be as specified herein and on Top Assembly TA13220E2899. The Toroid Assembly shall have the following electrical characteristics:

Resistance - $\leq$0.20 ohms
Voltage - 0.46 $\pm$ 0.05 Volts alterating current (Vac)

3.6.5 Input and filter assembly. The input and filter assembly shall be as specified herein and on Top Assembly TA13220E2898. The input and filter assembly shall exhibit the following impedance when measurements are made between input and transformer output terminals:

Resistance - (a)

Input/Output	Resistance
Terminal/Orange lead	$\leq$0.20 ohms
Terminal/White lead	$\leq$0.20 ohms

(b)

Input/Input	Resistance
Terminal/Terminal	$\geq$100K ohms

(c)

Output/Output	Resistance
Terminal/Terminal	$\geq$100K ohms

Voltage -

Generator Frequency and Voltage level	Output lead/dB tolerance
10 kilohertz (kHz) at 2.5 Vac	Orange/equal input, $\pm$1 dB
100 kHz at 2.5 Vac	Orange/decrease, 6 dB minimum
200 kHz at 2.5 Vac	Orange/decrease, 20 dB minimum
10 kHz at 2.5 Vac	White/equal input, $\pm$1 dB
100 kHz at 2.5 Vac	White/decrease, 6 dB minimum
200 kHz at 2.5 Vac	White/decrease, 20 dB minimum

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3.6.6 Circuit card assembly, signal processor. The signal processor circuit card assembly shall be in accordance with Top Assembly TA13220E2894. The signal processor circuit card assembly when installed in a MWS shall meet the characteristics of 3.6.1, 3.6.2, and 3.6.3.1 through 3.6.3.7.

3.7 Electromagnetic susceptibility. The electromagnetic susceptibility characteristics of the MWS shall conform to MIL-STD-461, Methods RS03, CS01, CS06, as modified in 3.7.1 and 3.7.2.

3.7.1 Radiated. Radiated susceptibility, Method RS03, shall be conducted in an electric field of 5 volts per meter (V/m) for both steady-state and keyed conditions. This field applies to frequencies up to 500 megahertz (MHz). Above 500 MHz the electric field shall be 1 V/m.

3.7.2 Conducted. Conducted interference signals, Method CS01 and CS06, shall be applied as follows:

(a) Continuous wave: 30 Hz to 400 kHz.
(b) Interference spike voltage: ten spikes, 5 to 10 ms wide, at a repetition rate of 7 to 10 pulses per second, on input leads only. Random phasing of the spikes with respect to power line frequency shall be employed.

3.8 Reliability. The MWS shall demonstrate a Mean-Time-Between-Failure (MTBF) of 10,000 hours when tested as specified in 4.6.2.4.

3.9 Identification marking. The MWS shall be identified in accordance with MIL-STD-130.

3.10 Treatment and painting. Unless otherwise specified (see 6.2), the portions of the MWS normally painted shall be cleaned, treated, and painted in accordance with MIL-C-22751, Lusterless Grey conforming to FED. STD. No. 595, Color No. 36440. Nonpermissible galvanic couples (dissimilar metals) as defined in MIL-T-704 shall be insulated as specified in MIL-T-704.

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3.11 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 6.5) for testing major components separately:

Item	Description	Identification	Quantity
1.	Sensor, Magnetic Weapon DT-547/FSS-9(V)	NSN 6350-00-228-2590 TA13220E2900	1

3.12 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustments, and without selection for fit or performance.

3.13 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9. These requirements are applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purpose intended.

3.13.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, Requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specification, standards, and drawings.

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4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of failure. Each disassembled part shall be examined in detail for compliance with the specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-loaned property shall be excluded from the requirements of this paragraph.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the supplementary quality assurance provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4.3 and 6.9).

4.2 Classification of inspections. The MWS shall be inspected as specified herein. Individual testing of major components of the complete MWS is not required. Major components procured separately shall be individually inspected as specified herein. Inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction MWS and major components (when procured separately) shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests.

4.3.2.1 Complete MWS. The complete MWS shall be subjected to the tests marked "X" in Column 1 of Table I. Individual testing of major components of the complete MWS is not required. Failure of any test shall be

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cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2.2 Major components procured separately. Signal processor assemblies when procured separately shall be subjected to the tests marked "X" in Column 1 of Table II. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2. The toroid assemblies and the input and filter assemblies (when procured separately), shall be tested as specified in 4.6.2.4.8 and 4.6.2.4.9 respectively. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Initial preduction inspection. When specified (see 3.3), one or more initial production MWS's will be selected at random by the Government from MWS's being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production MWS shall not exclude the remaining MWS's from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production MWS to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production MWS and shall be cause for refusal by the Government to continue acceptance of production MWS's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on MWS's previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all MWS's accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the defects specified in 4.6.1 AQL shall be 2.5 percent defective.

4.5.3 Tests.

4.5.3.1 Individual.

4.5.3.1.1 Complete MWS. Each complete MWS shall be subjected to the tests marked "X" in Column 3 of Table I. Individual testing of major components of the complete MWS is not required. Failure of any test shall be cause for rejection.

4.5.3.1.2 Major components procured separately. Each signal processor assembly (when procured separately), shall be subjected to the tests marked "X" in Column 2 of Table II. Failure of any test shall be cause for rejection.

4.5.3.2 Samples.

4.5.3.2.1 Complete MWS. Each complete MWS shall be subjected to the test marked "X" in Column 4 of Table I. Individual testing of major components of the complete MWS is not required. AQL shall be 2.5 percent defective.

4.5.3.2.2 Major components procured separately. Signal processor assemblies (when procured separately), shall be subjected to the tests marked "X" in Column 3 of Table II. The toroid assemblies and the input and filter assemblies (when procured separately), shall be tested as specified in 4.6.2.4.8 and 4.6.2.4.9, respectively. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The MWS and major components (when procured separately), shall be examined as specified herein for the following defects:

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101. Any part or component not in accordance with the SQAP requirements as shown on the drawings.
102. Components missing or not as specified.
103. Assembly not correct.
104. Electromagnetic interference suppression components and network not identical to the approved model.
105. Identification or special marking missing, incomplete or not legible.
106. Treatment and painting not as specified.
107. Interchangeability not as specified.
108. Workmanship including soldering not as specified.

4.6.2 Tests.

4.6.2.1 Test condition. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. Prior to testing, each MWS and signal processor card (when procured separately) shall be subjected to a 72 hour burn-in during which it shall be exercised through all its functions not less than nine times during each 24-hour period. All failures shall be recorded and analyzed for cause of failure. Corrective action shall be taken when appropriate. Copies of these failure reports shall be forwarded to the contracting officer. Units for testing shall be selected from those units having successfully completed the 72-hour burn-in.

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4.6.2.2 Test schedule.

TABLE I. Test schedule for complete MWS.

Preproduction	Initial production	Quality conformance: Individual	Quality conformance: Sample	Test	Requirement paragraph	Test paragraph
1	2	3	4	5	6	7
X	X	X	-	Functional:		
X	X	X	-	Power	3.6.3.1	4.6.2.4.1
X	X	X	-	Alarm sensitivity	3.6.3.2	4.6.2.4.2
X	X	X	-	Recovery	3.6.3.3	4.6.2.4.3
X	X	X	-	Circuit performance	3.6.3.4	4.6.2.4.4
X	X	-	X	Band-pass	3.6.3.5	4.6.2.4.5
X	X	-	X	Intermodulation	3.6.3.6	4.6.2.4.6
X	X	X	-	Tamper Switch tolerance	3.6.1	4.6.2.4.7
X	X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.1
X	X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.2.5.3
X	X	-	-	Shock	3.5.2(c)	4.6.2.5.4
X	X	-	-	Vibration	3.5.2(d)	4.6.2.5.5
X	X	-	-	Electromagnetic susceptibility	3.7	4.6.2.6
-	X	-	-	Reliability	3.8	4.6.2.7

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TABLE II. Test schedule for circuit card assembly, signal processor when procured separately. 1/

Preproduction	Quality conformance		Test	Requirement paragraph	Test paragraph
	Individual	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.6.6	4.6.2.4.10
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.3.2.1
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.3.2.2
X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.3.2.3
X	-	-	Shock	3.5.2(c)	4.6.3.2.4
X	-	-	Vibration	3.5.2(d)	4.6.3.2.5

1/Place the signal processor circuit card assembly in Government-loaned MWS or equivalent to simulate normal environment when installed in the MWS.

4.6.2.3 Calibration of test equipment. Calibration of test equipment shall be in accordance with MIL-C-45662.

4.6.2.4 Functional.

4.6.2.4.1 Power. The MWS shall be tested in accordance with Appendix Test Method 1, Procedure 2.1. Nonconformance to 3.6.3.1 shall constitute failure of this test.

4.6.2.4.2 Alarm sensitivity. The MWS shall be tested in accordance with Appendix Test Method 1, Procedure 2.1. Nonconformance to 3.6.3.2 shall constitute failure of this test.

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4.6.2.4.3 Recovery. The MWS shall be tested in accordance with Appendix Test Method 1, Procedure 2.1. Nonconformance to 3.6.3.3 shall constitute failure of this test.

4.6.2.4.4 Circuit performance. The MWS shall be tested in accordance with Appendix Test Method 1, Procedure 2.2. Nonconformance to 3.6.3.4 shall constitute failure of this test.

4.6.2.4.5 Band-pass. The MWS shall be tested in accordance with Appendix Test Method 1, Procedure 2.3. Nonconformance to 3.6.3.5 shall constitute failure of this test.

4.6.2.4.6 Intermodulation. The MWS shall be tested in accordance with Appendix Test Method 1, Procedure 2.4. Nonconformance to 3.6.3.6 shall constitute failure of this test.

4.6.2.4.7 Tamper switch tolerance. The MWS shall be tested in accordance with Appendix Test Method 2. Nonconformance to 3.6.1 shall constitute failure of this test.

4.6.2.4.8 Toroid Assembly. The toroid assembly (when procured separately) shall be tested in accordance with Top Assembly TA13220E2899. Nonconformance to 3.6.4 shall constitute failure of this test.

4.6.2.4.9 Input and Filter Assembly. The input and filter assembly (when procured separately) shall be tested in accordance with Top Assembly TA13220E2898. Nonconformance to 3.6.5 shall constitute failure of this test.

4.6.2.4.10 Circuit card assembly, signal processor. The signal processor circuit card assembly (when procured separately) shall be installed in the Government-loaned MWS and tested in accordance with 4.6.2.4.1 through 4.6.2.4.7. Nonconformance to 3.6.6 shall constitute failure of this test.

4.6.2.5 Environmental.

4.6.2.5.1 Low temperature. Conduct low temperature test in accordance with MIL-STD-810, Method 502.1, Procedure I, storage temperature -30° F, for not less than 48 hours, and operating temperature -20° F.

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4.6.2.5.2 High temperature. Conduct high temperature test in accordance with MIL-STD-810, Method 501.1, Procedure II, except that the nonoperating temperature shall be +165° F; and the operating temperature shall be +150° F.

4.6.2.5.3 Humidity. Conduct humidity test in accordance with MIL-STD-810, Method 507.1, Procedure I, with the following exceptions:

(a) Step 5 shall be for 5 cycles (not less than 120 hours). Measurements at the completion of each cycle.
(b) During Steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
(c) During Step 4, the relative humidity shall be maintained at 92 percent or greater.
(d) Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle. The cover of the MWS shall not be opened during measurements. The MWS may be removed from the environmental chamber for measurement; however, unsuccessful completion of measurements or unsuccessful return of the MWS to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of this test.

4.6.2.5.4 Shock.

(a) Basic design: In accordance with MIL-STD-810, Method 516.2, Procedure I, equipment not operating, sawtooth shock pulse, peak value 20g's duration 11 ms.
(b) Bench handling: In accordance with MIL-STD-810, Method 516.2, Procedure V.

4.6.2.5.5 Vibration. In accordance with MIL-STD-810, Method 514.2, Procedure X, Curve AX, equipment not operating.

4.6.2.5.6 Failure criteria. Nonconformance to 3.5 shall constitute failure of these tests.

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4.6.2.6 Electromagnetic interference.

4.6.2.6.1 Radiation susceptibility Test the MWS in accordance with MIL-STD-461, Method RS03. Electric field requirements of RS03 shall be modified and limited as specified in 3.7.1. All leads (sensor, alarm and power) external to the signal processor enclosure shall be exposed to the electric field. The leads shall be 4 or more feet in length.

4.6.2.6.2 Conducted susceptibility. Test the MWS in accordance with MIL-STD-461, Methods CS01 and CS06. Interference spike test specified in CS06 shall be modified, as specified in 3.7.2.

4.6.2.6.3 Failure criteria. The contractor shall furnish the contracting officer a test plan and a report of tests specified herein. Disapproval of the report shall constititute failure of this test.

4.6.2.7 Reliability. Using the uper MTBF (θ_0) of 10,000 hours and a lower test MTBF (θ_1) of 5,000 hours the MWS shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan VIIIC. Continue testing until either an "accept" or "reject" decision is reached. A failure if defined as any malfunction which causes or may cause one or more of the following:

(a) Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
(b) Serious damage to the system.
(c) Serious personnel safety hazards.

4.6.2.7.1 Test procedure. The tests shall be conducted under ambient laboratory conditions, without temperature, humidity or vibration cycling. Equipment shall be energized at the primary power input specified herein. Operation shall be continuous throughout the test with all circuits exercised not less than nine times during each 24-hour period. The parameters to be measured shall be in accordance with the appendix test methods. The tests shall commence after installation and burn-in period of 72 hours as specified in 4.6.2.1. Parts or components of known or limited life, such as batteries, shall have a life stipulated prior to test. Failures of these parts occurring after the end of the stipulated period are nonrelevant, but any dependent failures caused thereby are relevant. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the

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equipment test routine shall be reported and cause of the alarm indentified if practicable. This accounting shall be maintained separately from the failure accounting. Redundant elements, such as indicator lamps, shall not be repaired nor counted as failure until the MWS has failed to perform its intended function. Preventive maintenance procedures specified for the equipment during normal operation shall be performed during the reliability test. No additional preventive maintenance is allowed during the test.

4.6.2.7.2 Failure criteria. Nonconformance to 3.8 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select MWS's at any time during the contract production period and subject these MWS's the examination and the tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on MWS's selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production MWS's. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison MWS to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison MWS and shall be cause for refusal by the Government to continue acceptance of production MWS's until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on MWS's previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all MWS's accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

Major

109. Materials, methods and containers not as specified for Level A, or Level B. Each incorrect material, method, or container shall be considered one defect.
110. Boxed contents not cushioned to prevent movement or damage as specified for Level A or Level B.
111. Strapping not as specified for Level A.
112. Quantities packed together exceed the size or weight limitations of the box for Level A or Level B.
113. Marking missing, illegible, incorrect, or incompleted for Level A, Level B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A. The MWS components shall be preserved by the applicable following procedure, depending upon procurement of complete MWS's or procurement of components separately.

5.1.1.1 Procurement of complete MWS's. The signal processor circuit card assembly contained within the signal processor enclosure, and the toroid assembly (end of line module) shall each be preserved in accordance with MIL-P-116, Method IIc. The loop wire shall not require proservation. The components shall then be placed together in a close-fitting box conforming to PPP-B-636, W6c, style optional. The components shall be cushioned within the box to prevent movement or damage.

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5.1.1.2 Major components furnished separately. The loop wire shall not require preservation. All other components shall be individually preserved in accordnace with MIL-P-116, Method IIc. Each component shall then be placed in a close-fitting box conforming to PPP-B-566, Variety 1, style, type, and class optional or PPP-B-676, Variety 1, style, type, and class optional. The components shall be cushioned to prevent movement or damage within the box.

5.1.1.2.1 Consolidation. MWS components of like description, preserved as specified in 5.1.1.2, shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c or W5c, style optional, in quantities not to exceed the weight limitation of the box. The contents shall be cushioned within the box to prevent movement or damage.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for Level A, except Method IA-8 shall be used in place of Method IIc.

5.1.3 Commercial. Each complete MWS and MWS component procured separately shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B or Commercial as specified (see 6.2).

5.2.1 Level A. The MWS components, preserved as specified in 5.1, shall be packed by the applicable following procedure depending upon procurement of complete MWS's or procurement of MWS components separately.

5.2.1.1 Packing of complete MWS's. Complete MWS's shall be packed together in a close-fitting box conforming to PPP-B-601, Overseas Type Grade B, style optional, or PPP-B-621, Class 2, style optional, in quantities not to exceed the weight limitation of the box. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.1.2 Packing of major components procured separately. MWS components procured separately, preserved as specified in 5.1, shall be packed together as specified in 5.2.1.1. Only components of like description shall be packed together.

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5.2.2 Level B. Complete MWS's and MWS components procured separately, shall be packed as specified in 5.2.1.1 or 5.2.1.2 as applicable except boxes shall be Domestic Type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Complete MWS's and MWS components procured separately, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.3.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188 and the special marking specified in 5.3.3.

5.3.3 Special marking. Each shipping container shall be marked with code labels in accordance with MIL-STD-129. The labels shall contain the Project Code "JID".

6. NOTES

6.1 Intended use. The MWS is intended for use as a component of the J-SIIDS. The MWS is used to protect arms rooms at Government installations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When major components are to be furnished separately and quantity to be furnished (see 3.1.1).
(c) Time frame required for submission of preproduction model and number of MWS's and major components (when procured separately) required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examinations and tests. When the Government will conduct some but not all of the preproduction examinations and tests, the contracting officer should specify which

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examinations and tests will be conducted by the Government and which examinations and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required, and number of MWS's to be furnished, (see 3.3).
(f) When cleaning, treating, painting, and the color shall be other than as specified (see 3.10).
(g) When Government-loaned property is not required of quantity is other than specified (see 3.11).
(h) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(i) When other than Finish B strapping is required (see 5.2.1.1).

6.3 Preproduction model. Any changes or deviations of production MWS's or major components (when procured separately) from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish MWS's (or major components (when procured separately) conforming to this specification.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the MWS is designed to alarm.

6.4.2 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, change of fit of any component, physical change which impairs the mechanical integrity of the MWS, evidence of delamination or water penetration into integrated circuits, printed circuit boards, or components.

6.4.3 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. A SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of the repair parts.

6.5 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.11, when applicable (see 6.2).

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6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each MWS.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of MWS's.

6.8 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

6.9 Quality assurance records. The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected and corrective action taken when applicable."

Custodians:
Army - ME
Air Force - 99

Review activity
DLA - GS

Preparing activity:
Army - ME

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APPENDIX

TEST METHODS

FOR

SENSOR, MAGNETIC WEAPON

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether Magnetic Weapon Sensors and Signal Processor Circuit Assemblies procured under this specification conform to the requirements. The Signal Processor Circuit Assembly shall be installed in a Government-loaned MWS for this test.

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TEST METHOD 1

MWS FUNCTIONAL TEST

1. TEST APPARATUS

(a) Power supply, 20 ±2 Vdc.
(b) Signal generator, battery powered.
(c) Voltohmmeter.
(d) RMS voltmeter.
(e) Special attenuator - See Figure 1.

2. TEST PROCEDURE

CAUTION

Enclosure terminal (TB1-9) should be grounded to avoid shock hazards.

2.1 Power, alarm sensitivity and recovery test.

(a) Apply reverse voltage of 20 Vdc to TB1-7 and TB1-8. There shall be no damage.
(b) Turn the MWS sensitivity control (R35) to Position 1 (minimum gain). Place unit in operation by applying 20 Vdc to TB1-7 and TB1-8. With toroid assembly, or electrical equivalent, connected to input and filter assembly terminals. There shall be no alarm at TB1-1 and TB-2, or tamper alarm at output TB1-5 and TB1-6. Output (no alarm) resistance shall be 10 ohms or less.
(c) Interrupt 20 Vdc input power for at least 0.5 seconds. Output (alarm) resistance shall be greater than 100k ohms.
(d) Short the loop wire input momentarily. The MWS shall alarm for 0.5 to 5.0 seconds.
(e) Connect a 3 ohm resistor across the input and wait 15 seconds. Output shall be no alarm. Then, using the attenuator in Figure 1, supply 100 nanovolt peak positive signal of 0.65 Hz across the resistor. The MWS shall alarm.
(f) Repeat step (e) with negative signal.
(g) Measure the current at TB1-7 with tamper and alarm activated. Current shall not exceed 60 mA.

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(h) Turn the MWS sensitivity control (R35) to Position 6 (maximum gain). Repeat steps (d) through (g) with a 25 nanovolt signal.

2.2 Circuit performance test.

(a) Place a 3 ohm resistor across the input terminals (omit input signal sources). Set R35 at Position 1, (minimum gain).
(b) The voltage at the output of A2 (A2-6) shall be less than 46 millivolts, indicating that the input voltage is less than 4 nanovolts (4×10^{-9}) for an observed period of 60 seconds.

2.3 Band-pass test.

(a) Set the sensitivity control to Position 1 (minimum gain) and apply a 115 nanovolts (0.35 Hz) signal across the 3 ohm resistor.
(b) Wait 15 seconds, then average the peak value of the next 10 cycles at the output of A2 (A2-6). The average shall be between 0.56 volts and 0.89 volts peak.
(c) Apply a 25 nanovolt (1.3 Hz) peak signal across the 3 ohm resitor at the input terminals.
(d) Wait 15 seconds, then average the peak value of the next 10 cycles at the output of A2 (A2-6). The average shall be between 0.57 volt and 0.91 volt peak.
(e) Apply a 5120 nanovolt (10 Hz) peak signal across the 3 ohm resistor at the input terminals.
(f) Wait 15 seconds, then average the peak value of the next 10 cycles at the output of A2 (A2-6). The average shall be 0.91 volts peak maximum.
(g) Apply a 910 nanovolt (0.1 Hz) peak signal across the 3 ohm resistor at the input terminals.
(h) Wait 15 seconds, then average the peak value of the next 10 cycles at the output of A2 (A2-6). The average shall be 0.91 volts peak maximum.

2.4 Intermodulation test.

(a) Apply both a 150 microvolt (30 Hz) signal and a 150 microvolt (32 Hz) signal across the 3 ohm resistor at the input terminals, simultaneously. Wait 15 seconds, then monitor for an additional 6 seconds. The MWS shall not alarm.

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(b) Apply both a 350 microvolt (rms) (62 Hz) signal and a 350 microvolt (rms) (60 Hz) signal across the 3 ohm resistor at the input terminals, simultaneously. Wait 15 seconds, then monitor for an additional 6 seconds. The MWS shall not alarm.

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TEST METHOD 2

TAMPER SWITCH TOLERANCE TEST

1. TEST APPARATUS

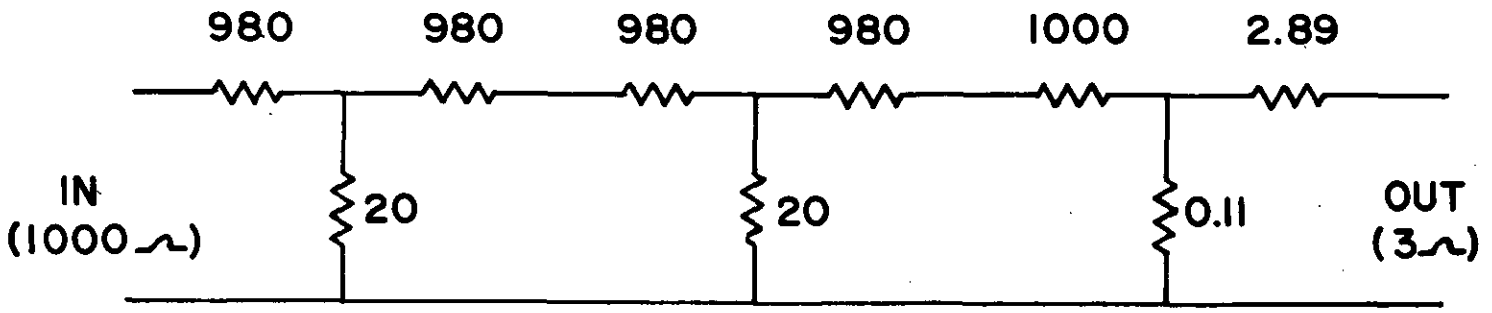
(a) Power supply, 20 ±2 Vdc.
(b) Signal generator.
(c) Voltohmmeter.

2. TEST PROCEDURE

2.1 Signal processor. Place the MWS in operation by applying 20 Vdc to Terminals TB1-7 and TB1-8 of the signal processor.

(a) Connect an ohmmeter between the tamper terminals on the signal processor TB1-5 and TB1-6.
(b) Close the cover of the signal processor enclosure and allow the signal processor to settle in the no-alarm state. The resistance across the tamper terminals (TB1-5 and TB1-6) shall be 10 ohms or less.
(c) The resistance between TB1-5 and TB1-6 shall increase to greater than 100k ohms when the cover is raised more than 0.062 inch but not more than 0.187 inch. Alarm.
(d) The resistance between TB1-5 and TB1-6 shall be greater than 100k ohms when the cover is opened. Alarm.
(e) Pull the tamper switch fully out and lock. The resistance between TB1-5 and TB1-6 shall decrease to 10 ohms or less. No alarm.
(f) Repeat steps (b) and (c).

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ATTENUATION 40db + 40db + 60db = 140db TOTAL

NOTE:
ALL VALUES OF RESISTANCE IN OHMS. CALIBRATE TO CONFIRM Z OUT = 3 ± 0.05 OHMS AND THAT TOTAL ATTENUATION = 140 ± 0.2db

FIGURE 1. Attenuator.

X-3056

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MIL-S-52866B Sensor, Magnetic Weapon DT-547/FSS-9(V)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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